

Object Transportation by Cooperative Mobile Manipulators in Coordination with Humans

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ABSTRACT

This paper introduces a real-time motion planning framework using Nonlinear Model Predictive Control (NMPC) for a group of mobile manipulators carrying a load together in coordination with humans. The described method allows for safe and robust navigation in dynamic environments by optimizing several objectives simultaneously. The key to our framework is a new dynamic coordinated approach that enables successful task completion by the human-robot team. The constraints of the system, such as robot kinematics, rigid formation maintenance, and collision avoidance among the robots as well as with the obstacles achieved using Velocity Obstacle (VO) are incorporated directly in its NMPC. The robot team actively coordinates with the human partner by issuing signals to modulate their speed or pause the task in response to predicted task difficulty and safety-critical thresholds in complex, cluttered environments. The efficacy of the proposed method is validated across over 50 simulation trials in high-fidelity *CoppeliaSim* environments, achieving a 96% task completion rate compared to 18.2% for a baseline controller without coordination, demonstrating safe object transport, rigid formation maintenance, and robust human-partner tracking in cluttered environments.

1. Introduction

Usage of cooperative teams of multiple mobile manipulators represents significant advancement in automation, providing capabilities such as moving large objects, fixture-less assembly, and logistics in large-scale settings [1]. Through the integration of the mobility of mobile platforms with the dexterity of robotic arms, these systems provide increased flexibility and payload as compared to an individual robot. However, to unlock this potential, advanced motion planning and control strategies are required to facilitate multi-robot coordination, preserve formation structure, and ensure safety in dynamic, unstructured environments [2, 3, 4].

To meet these challenges, optimization-based control techniques, especially Model Predictive Control (MPC), have proven to be a strong and highly appropriate paradigm. Predictive control in MPC enables the system to look forward and plan optimal control moves over a finite horizon while actively considering system dynamics, control boundaries, and environmental limits for holonomic and non-holonomic robots [5]. MPC has been effectively used by researchers for cooperative transport to generate smooth and collision-free trajectories for the robot formations [6, 7].

In an increasingly common range of tasks, ranging from factory floors to warehouse operations, robots must work not independently but in close coordination with human partners. In these cases, the desired path is not known in advance but rather generated in real-time by a human leader. This poses a human-in-the-loop control problem with

three distinct challenges: Firstly, the human's trajectory is a priori unknown to the robot system and potentially non-smooth. Secondly, the robotic team is constrained by its own kinodynamic and physical limitations, which can prevent it from matching the human's agility. Thirdly, a naive reactive controller can suffer from oscillations or instability, as the inherent latency in the robots' response can lead to undesirable fluctuations in the human-robot distance.

The challenge of human-initiated cooperative transport is at the intersection of multiple areas of research. Cooperative transportation of objects has been widely researched [8, 9], with solutions from distributed control schemes [10] that enhance scalability. In the field of human-robot interaction, "human-aware motion planning" research tends to concentrate on safety and legibility, making robot motions predictable for proximal humans [11, 12]. Lastly, leader-follower formation control gives the basic control laws for preserving geometric structures [13], yet usually considers an ideal robotic or virtual leader with known dynamics.

The motivation for developing MPC-based human-follower control of cooperative mobile manipulators arises from several critical research needs and industry demands for applicability. While existing cooperative manipulation systems have proved to be effective in controlled spaces with pre-defined motions, real-world deployments demand human-robot collaboration with the capability for dynamic human guidance of robot teams in complex tasks. Traditional formation control techniques [13] do not include inherent variability and uncertainty of human gait patterns.

Current distributed control systems [10, 14], though scalable, are not predictive and thus cannot fill the temporal gap between human intention and robot's ability to execute.

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The temporal gap typically results in jerky motions, formation degradations, and potential safety concerns in collaborative environments. Further, current human-aware motion planning methods [11, 15] are primarily designed for use in single-robot operation and fail to completely address the advanced coordination requirements of multi-robot teams guided by human leaders.

Modern manufacturing and logistics require adaptive automation technology with the capability to change with evolving production requirements and operate in coordination with human operators. Having a human operator naturally guide a group of mobile manipulators for processes like large assembly operations, user-configurable manufacturing layouts, and dynamic material handling is a leap ahead in human-robot collaboration. These systems have the ability to apply human mental capabilities to higher-level decision making and robotic power and accuracy for physical handling work.

In contrast to reactive controllers, MPC's predictive capability permits the system to anticipate motion from humans and design coordinated maneuvers that preserve formation integrity while providing smooth tracking performance. The optimization paradigm inherent in MPC allows multiple objectives, such as human tracking, formation preservation, collision prevention, and dynamic satisfaction of constraints, to be addressed simultaneously [4, 16]. This ability to optimize is particularly critical in dealing with the competing demands of following a human leader while maintaining safety and effectiveness in cooperative manipulation. With MPC, since it can re-plan on a continuous basis using updated information, the uncertain and dynamic aspects of human-paced collaborative tasks can be handled more effectively.

Our method incorporates the human-robot interaction into the control loop formally by proposing a new coordination approach that keeps the team together.

The primary contributions of this paper are threefold:

1. A unified MPC framework that simultaneously optimizes for human-guided trajectory tracking, rigid formation maintenance, and multiple dynamic obstacle collision avoidance.
2. A novel, multi-level adaptive coordination protocol that ensures robust human-robot coupling by giving feedback to human partner based on both predictive performance metrics and reactive safety thresholds.
3. Validation of the integrated system in a high-fidelity simulation with both static and dynamic obstacles, demonstrating safe and fluid navigation.

The subsequent sections detail the related work (2), system model (3), the proposed MPC control framework(4), and the adaptive coordination protocol(5), followed by a rigorous experimental validation(6) and concluding remarks(7).

2. Related Work

While multi-robot transportation and human-robot interaction have matured as independent fields, their seamless integration remains an evolving research frontier. The following subsections reviews how current research addresses predictive optimization, coordination stability, and the technical strategies employed to bridge the temporal gap between human intent and robotic execution.

2.1. MPC for Cooperative Payload Transport

In cooperative manipulation, the primary challenge is managing the coupled dynamics of the mobile platforms and the robotic arms while maintaining a firm grasp on a shared object. To address this for heterogeneous teams, Lissandrini et al. [17] proposed a decentralized leader-follower NMPC scheme for cooperative payload transport. Their framework utilizes an elastic joint model to minimize internal forces and torques that could damage the payload, while relying on first-order kinematics to ensure robustness against model uncertainties. In contrast, Nikou et al. [18] developed an NMPC framework for multi-agent systems that explicitly accounts for full second-order dynamics and rigid grasp constraints. While their approach ensures collision and singularity avoidance during navigation to a final pose, both frameworks primarily assume static set-points or pre-calculated trajectories. This creates a significant gap in human-guided navigation scenarios, where the target state is stochastic and must be updated at every sampling interval based on real-time human intent.

2.2. Human-Leader Formation Dynamics

When a human guides a robotic team, the system must interpret kinodynamic constraints alongside the unpredictable velocity and trajectory of the human. To address the complexity of joint trajectory generation, Masone et al. [19] proposed a bilateral shared framework where a human supervisor and an autonomous algorithm jointly control a dynamic B-spline path. Their approach utilizes integral haptic feedback to inform the human operator of any mismatch between the ideal reference path and the actual traveled path, ensuring safety and collision avoidance while the human steers the robot. Complementing this shared steering approach, Wang et al. [20] developed a whole-body control framework for mobile manipulators that allows a human to guide the system through direct physical traction. By employing variable-damping admittance control, they ensure smooth movement during transportation and increased stability when the human signals a stop. To further enhance predictive fluid movement, Dani et al. [21] integrated human-intent observers (such as IMM and EM algorithms) with Control Barrier Functions (CBFs). This combination allows the robot to forecast the human's reaching actions and proactively generate safe trajectories that strictly avoid the operators workspace while maintaining synchronization. Our work extends these single-robot shared planning and intent-aware paradigms to multi-robot team formations,

where the predictive MPC must balance rigid payload transport constraints with the unpredictable trajectories inherent in human-guided motion.

2.3. Latency and Adaptive Coordination Protocols

A significant issue in human-follower systems is the potential for unstable or oscillatory responses caused by the physical limits of robots attempting to match human agility. Duchaine et al. [22] identified that while stability observers are necessary for safety, overly conservative control often degrades "transparency," making the robot a physical burden that resists the human's intuitive motion. Their framework specifically accounts for time delays (T) within the stability observer to prevent these unstable behaviors. To address the inherent instability caused by feedback delays, Lee and Huang [23] established the Passive Set-Position Modulation (PSPM) framework, which mathematically guarantees system passivity and stability even under significant communication latency and packet loss. Complementing this, Masone et al. [19] also addressed latency through haptic mismatch feedback, informing the leader when their pace exceeds the robot's feasible trajectory. Our work extends these adaptive protocols to multi-robot transport, providing real-time predictive performance metrics to the human partner to prevent formation breakdown under stochastic guidance.

2.4. Collision Avoidance in Constrained Formations

Navigating a multi-robot team through obstacles while transporting a shared payload significantly restricts the available solution space for MPC, as the system must treat the ensemble-payload unit as a single constrained entity. To address this, Marino [24] proposed a distributed architecture for networked mobile manipulators that utilizes the null-space of the Jacobian to achieve secondary objectives, such as obstacle avoidance, without compromising the primary formation task or increasing internal stresses on the shared object. For formations requiring rigid shape maintenance, Nfaileh et al. [25] demonstrated that integrating Virtual Structures with nonlinear MPC and artificial potential fields allows teams to maintain integrity even under actuator saturation and environmental disturbances.

To handle unstructured environments populated by humans, Brito et al. [26] utilized Model Predictive Contouring Control (MPCC) to compute trajectories within convex free-space regions. Their method proactively avoids dynamic obstacles by modeling them as ellipsoids and calculating safety boundaries based on the Minkowski sum of the obstacle and robot geometry. These predictive strategies are further supported by autonomous replanning strategies to escape local minima and Control Barrier Functions (CBFs) to enforce strict safety sets. Our work builds on these foundations by adapting optimization-based null-space projections to preserve rigid formation integrity under uncertain human-guided motion.

3. System Modeling and Problem Definition

We consider a system of N cooperative mobile manipulator robots (MMRs) that grasp a rigid object to transport it in coordination with a human within an environment containing static and dynamic obstacles. We define a stationary world frame, denoted as W , to serve as our universal point of reference. The object itself has its own reference frame, O , which has been fixed to its center of mass (COM). Similarly, every individual mobile manipulator, indexed by i , has its own local body frame, $\{B_i\}$, centered on its mobile base. Unless specified otherwise, all quantities are defined in the world frame $\{W\}$. The reference frame relationships and robot configurations are illustrated in Fig. 1.

3.1. Discrete-Time Kinematics of the Holonomic Fleet

The state of the i -th mobile manipulator, $\mathbf{z}_i \in \mathbb{R}^{3+\text{dof}}$, is a composite of its base position, $\mathbf{p}_i = [x_i, y_i, \theta_i]^T \in \text{SE}(2)$, and the vector of its manipulator joint angles, $\mathbf{q}_{a,i} \in \mathbb{R}^{\text{dof}}$. Putting it all together, the full state vector $\mathbf{z}_i$ for any robot i is defined by concatenation of base and arm states

$$\mathbf{z}_i = [\mathbf{p}_i^T, \mathbf{q}_{a,i}^T]^T$$

The system is controlled by commanding the base's linear velocities in the world frame, $\mathbf{u}_{b,i} = [v_{x,i}, v_{y,i}, \omega_i]^T$, and the arm's joint velocities, $\mathbf{u}_{a,i} = \dot{\mathbf{q}}_{a,i}$. The full control vector for the i -th robot is

$$\mathbf{u}_i = [\mathbf{u}_{b,i}^T, \mathbf{u}_{a,i}^T]^T.$$

The continuous-time dynamics of the i -th robot are governed by:

$$\dot{\mathbf{p}}_i(t) = \mathbf{u}_{b,i}(t) \quad (1)$$

$$\dot{\mathbf{q}}_{a,i}(t) = \mathbf{u}_{a,i}(t) \quad (2)$$

For implementation within a digital control framework, we discretize the system dynamics using a first-order Euler approximation with a sampling time Δt . The discrete-time state $\mathbf{z}_i[k]$ at time step k evolves to the next state $\mathbf{z}_i[k+1]$ according to the state transition function f :

$$\mathbf{z}_i[k+1] = f(\mathbf{z}_i[k], \mathbf{u}_i[k]) \quad (3)$$

where the function f represents the discrete kinematic model:

$$\mathbf{p}_i[k+1] = \mathbf{p}_i[k] + \Delta t \cdot \mathbf{u}_{b,i}[k] \quad (4)$$

$$\mathbf{q}_{a,i}[k+1] = \mathbf{q}_{a,i}[k] + \Delta t \cdot \mathbf{u}_{a,i}[k] \quad (5)$$

The complete state of the robot fleet is the concatenation of individual robot states, $\mathbf{Z}[k] = [\mathbf{z}_1[k]^T, \dots, \mathbf{z}_N[k]^T]^T$, and the collective control input is $\mathbf{U}[k] = [\mathbf{u}_1[k]^T, \dots, \mathbf{u}_N[k]^T]^T$. The motion of the system is subject to physical constraints. The sets of admissible states and controls are defined by the following bounds for each robot i at any time step k :

$$\|\mathbf{u}_{b,i}[k]\|_2 \leq V_{\max} \quad (6)$$

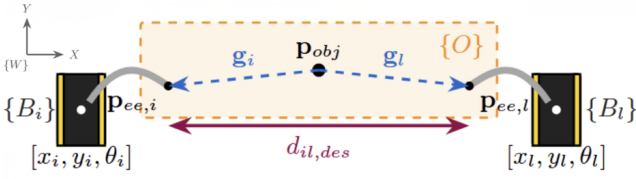


Figure 1: Geometric configuration and reference frame definitions for the cooperative transport system. The world frame $\{W\}$, object frame $\{O\}$, and local robot body frames $\{B_i, B_l\}$ provide the basis for the kinematic transformations. The diagram illustrates the relationship between the robot states $[x, y, \theta]$, end-effector positions $\mathbf{p}_{ee}$, and the object's center of mass $\mathbf{p}_{obj}$ via the fixed grasp offsets $\mathbf{g}_i$ and $\mathbf{g}_l$. The desired formation rigidity is maintained by enforcing the constant distance $d_{il,des}$.

$$\mathbf{q}_{a,i,\min} \leq \mathbf{q}_{a,i}[k] \leq \mathbf{q}_{a,i,\max} \quad (7)$$

$$\mathbf{u}_{a,i,\min} \leq \mathbf{u}_{a,i}[k] \leq \mathbf{u}_{a,i,\max} \quad (8)$$

where $V_{\max}$ is the maximum permissible speed of the mobile base, and the subscripts $(\cdot)_{\min}$ and $(\cdot)_{\max}$ on the variables $\mathbf{q}_{a,i}$ and $\mathbf{u}_{a,i}$ denote their minimum and maximum physical limits, respectively. These constraints define the feasible operating space for the optimization problem.

3.2. Cooperative Transport and Formation Model

The N robots rigidly grasp a planar object, forming a single composite rigid body, refer Fig. 1. The position of the center of mass of the object, denoted $\mathbf{p}_{obj} \in \mathbb{R}^2$, is the primary state variable for our control objective, where the NMPC generates robot commands to minimize the tracking error between $\mathbf{p}_{obj}$ and the dynamic target. The end-effector pose of each manipulator is determined from the robot's state using the Product of Exponentials (PoE) formula, which provides a complete kinematic description [27], [28].

The forward kinematics map for the i -th manipulator, which computes the end-effector's homogeneous transformation matrix $\mathbf{T}_{ee,i} \in SE(3)$ from the robot's state $\mathbf{z}_i$, is given by:

$$\mathbf{T}_{ee,i}(\mathbf{z}_i) = \mathbf{T}_{base,i}(\mathbf{p}_i) \left(\prod_{j=1}^{\text{dof}} e^{[S_j]q_{a,i,j}} \right) \mathbf{M}_i \quad (9)$$

where:

- The transformation for the mobile base, $\mathbf{T}_{base,i}(\mathbf{p}_i)$, maps the base's 2D pose $\mathbf{p}_i = [x_i, y_i, \theta_i]^T$ into $SE(3)$, assuming a constant height $z = 0$. It is constructed as:

$$\mathbf{T}_{base,i}(\mathbf{p}_i) = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & x_i \\ \sin(\theta_i) & \cos(\theta_i) & 0 & y_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- $S_j \in \mathbb{R}^6$ is the screw axis for the j -th joint of the manipulator arm.

- $q_{a,i,j}$ is the angle of the j -th joint of the i -th manipulator.
- $e^{[S_j]q_{a,i,j}}$ is the matrix exponential of the screw axis, representing the transformation of the j -th joint.
- $\mathbf{M}_i$ is the end-effector's home configuration (the pose of the end-effector when all joint angles are zero) relative to the manipulator's base.

The homogeneous transformation matrix $\mathbf{T}_{ee,i} \in SE(3)$ is a 4×4 matrix that compactly represents both the orientation and position of the end-effector. This matrix has the block structure:

$$\mathbf{T}_{ee,i} = \begin{bmatrix} \mathbf{R}_{ee,i} & \mathbf{p}_{ee,i} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}$$

where $\mathbf{R}_{ee,i} \in SO(3)$ is the 3×3 rotation matrix defining the end-effector's orientation, and $\mathbf{p}_{ee,i} \in \mathbb{R}^3$ is the position vector of its origin in the world frame. Therefore, the end-effector position $\mathbf{p}_{ee,i}$ is extracted by selecting the first three elements of the fourth column of the $\mathbf{T}_{ee,i}$ matrix.

The position of the object's COM, $\mathbf{p}_{obj}$, is kinematically related to the state of any robot i in the formation. The object's orientation in the world frame, represented by the rotation matrix $\mathbf{R}(\psi[k])$, is kinematically coupled to the orientation of any end-effector due to the rigid grasp. It can be computed from the rotational part of the end-effector's pose, $\mathbf{R}_{ee,i}[k]$, which is extracted from the homogeneous transformation matrix $\mathbf{T}_{ee,i}[k]$ computed in (9). Assuming the end-effector frame is aligned with the object's grasp frame, we have $\mathbf{R}(\psi[k]) = \mathbf{R}_{ee,i}[k]$. Given a fixed grasp point $\mathbf{g}_i$ on the object (in the object's local frame), the world-frame position of the object's COM can be derived from the i -th robot's end-effector position:

$$\mathbf{p}_{obj}[k] = \mathbf{p}_{ee,i}[k] - \mathbf{R}(\psi[k])\mathbf{g}_i \quad (10)$$

where $\psi[k]$ is the object's orientation and $\mathbf{R}(\psi[k])$ is the corresponding rotation matrix.

The rigid grasp imposes a set of holonomic equality constraints on the system. These constraints dictate that the squared Euclidean distance between the end-effectors of any pair of robots (i, l) must remain constant:

$$\|\mathbf{p}_{ee,i}[k] - \mathbf{p}_{ee,l}[k]\|_2^2 - d_{il,des}^2 = 0, \quad \forall i, l \in \{1, \dots, N\}, i \neq l \quad (11)$$

where $\mathbf{p}_{ee,i}[k]$ is computed via (9), and $d_{il,des}$ is the fixed, desired distance between the grasp points on the object. This constraint ensures the geometric integrity of the formation throughout the task.

3.3. Environment Perception

The operational workspace contains a set of static obstacles, $\mathcal{O}_s$, and dynamic obstacles, $\mathcal{O}_d$. We make the following perception assumptions:

- The geometric properties and locations of all static obstacles are known a priori from a pre-loaded map.

- Dynamic obstacles are tracked in real-time by the formation's sensors, providing their current position and velocity to the motion planner.

To ensure safe navigation, collision avoidance is enforced at the level of individual robots, not just the formation as a whole. Each robot i and each obstacle m is modeled with a conservative bounding volume (e.g., a circle), denoted $C_{\text{robot},i}$ and $C_{\text{obs},m}$, respectively. The fundamental safety requirement dictates that these volumes must never overlap. This is formalized by requiring the distance between volumes to remain above a safe margin:

$$\text{dist}(C_{\text{robot},i}(t), C_{\text{obs},m}(t)) \geq d_{\text{safe}}, \quad \forall i, m \quad (12)$$

To address the distinct challenges posed by static and dynamic environments, the collision avoidance strategy is bifurcated within the NMPC framework (detailed in Section (4)) :

- For static obstacles, the distance-based constraint from Eq. (12) is applied directly.
- For dynamic obstacles, a more sophisticated predictive method, the Velocity Obstacle (VO) technique, is employed to proactively compute collision-free maneuvers.

This multi-layered approach provides a robust safety guarantee for navigation in complex, cluttered environments.

3.4. Human Perception System

To enable robust tracking of the human partner, we assume the robot team is equipped with a perception system capable of estimating the human's position in real-time. This can be achieved through a variety of line-of-sight (LOS) sensing modalities (e.g., LiDAR or depth camera).

The sensor provides measurements of the human's position relative to the robot's own frame. These relative measurements are then transformed into the world frame to produce a globally consistent estimate of the human's position, $\mathbf{p}_h[k]$.

A critical characteristic of such LOS sensors is a finite perception range, which we denote as the *Visual Sensor Limit* (fixed at 5.5 m). If the human-robot separation distance exceeds this maximum sensor range, the perception link is broken, and the system can no longer obtain an estimate of the human's position. This practical limitation is a key factor in our experimental validation, as detailed in Section 6.2.3.

The estimated position, $\mathbf{p}_h[k]$, serves as the primary input for generating the reference target. To define a target point, we first estimate the human's direction of motion, $\mathbf{d}_h[k]$, from a short history of their recent positions:

$$\mathbf{v}_h^{\text{est}}[k] = \frac{\mathbf{p}_h[k] - \mathbf{p}_h[k-1]}{\Delta t} \quad (13)$$

$$\mathbf{d}_h[k] = \frac{\mathbf{v}_h^{\text{est}}[k]}{\|\mathbf{v}_h^{\text{est}}[k]\|_2 + \epsilon_v} \quad (14)$$

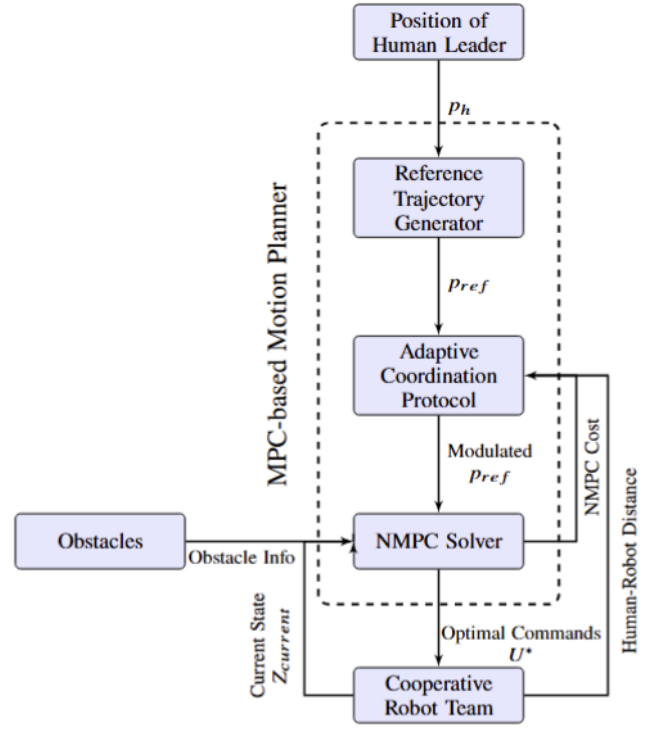


Figure 2: Architecture flow of the proposed MPC-based motion planner.

where ϵ_v is a small constant to prevent division by zero. The reference target point, $\mathbf{p}_{\text{ref_target}}[k]$, is then defined at a safe standoff distance, D_{standoff} , directly behind the human partner:

$$\mathbf{p}_{\text{ref_target}}[k] = \mathbf{p}_h[k] - D_{\text{standoff}} \cdot \mathbf{d}_h[k] \quad (15)$$

This target serves as the instantaneous goal for the MPC, as detailed in Section (4) .

3.5. Dynamic Reference Generation

A fundamental challenge in human-guided transport is that the human's future trajectory is unknown. Predictive models based on velocity can be unreliable, as human intent can change abruptly. To address this, we propose a novel framework where the reference trajectory for the MPC is dynamically re-generated at every time step based on the human's current position, refer Fig.(3). This is coupled with a two-level feedback protocol that uses the outcome of the MPC solve to implicitly communicate the robot team's capabilities to the human.

Horizon-Specific Reference Trajectory Generation

The only information available to the robot team is the human's current position, $\mathbf{p}_h[k]$, which is obtained via the perception system. At the beginning of each control cycle at time k , we define a single target point for the horizon, $\mathbf{p}_{\text{ref_target}}[k]$, located at a standoff distance behind the human's current position (calculated using Eq. 15).

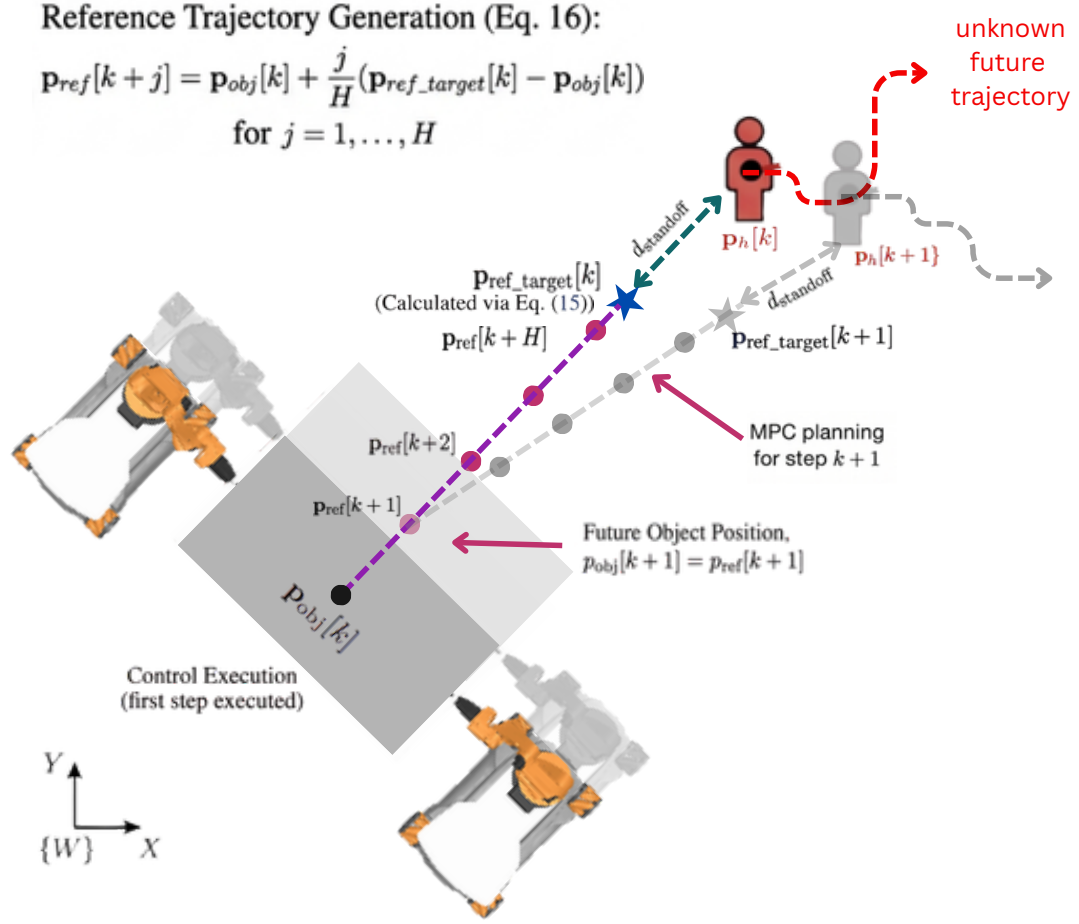


Figure 3: Dynamic Reference Trajectory Generation (Eq. 16). The figure illustrates the MPC reference trajectory generation for human-guided transport at time step k . The human's current position ($\mathbf{p}_h[k]$) dictates the horizon-specific target point ($\mathbf{p}_{ref_target}[k]$) at a standoff distance $d_{standoff}$. The reference trajectory (purple dashed line), defined by Eq. (16), spans from the object's current position ($\mathbf{p}_{obj}[k]$) to $\mathbf{p}_{ref_target}[k]$ over horizon H . The ghosted elements (grey) represent the state at a subsequent time step, $\mathbf{p}_h[k+1]$ and its corresponding reference, demonstrating the dynamic re-generation of the trajectory based on the human's continuous movement. The "MPC planning for step $k+1$ " annotation highlights the iterative planning, where $\mathbf{p}_{obj}[k+1]$ is the expected object position after executing the first step of the current plan.

Instead of predicting the human's future path, we create a simple, straight-line reference trajectory that serves as the MPC's goal for the current planning horizon, H . This trajectory starts at the formation's current center of mass, $\mathbf{p}_{obj}[k]$, and ends at the target point, $\mathbf{p}_{ref_target}[k]$. The reference point for each step j within the horizon is calculated via linear interpolation:

$$\mathbf{p}_{ref}[k+j] = \mathbf{p}_{obj}[k] + \frac{j}{H} \cdot (\mathbf{p}_{ref_target}[k] - \mathbf{p}_{obj}[k]), \quad (16)$$

for $j = 1, \dots, H$

This strategy effectively commands the MPC to find the best possible path to "catch up" to the human's current position over the next H steps. As the human moves, this target is updated at every cycle, leading to a continuously adapting, emergent following behavior.

3.6. Human-Guided Trajectory Following

The primary task objective is to follow a human partner, whose position is denoted by $\mathbf{p}_h[k]$. The control objective is to generate commands that steer the object's Center of Mass (COM), $\mathbf{p}_{obj}[k]$, towards a dynamically generated reference target, which is located at a safe standoff distance behind the human's perceived position.

However, simply following this reference is insufficient for robust human-robot collaboration. The robotic team is constrained by its own kinodynamic and physical limitations, which can prevent it from matching the human's agility.

To address this, a critical component of our problem is to establish an adaptive coordination mechanism that creates a feedback loop from the robots to the human collaborator. This mechanism operates on two distinct levels:

- **A proactive level** that uses the NMPC's predictive cost to anticipate task difficulty and signals the human partner to moderate their speed, thereby ensuring smooth, continuous tracking.
- **A reactive level** that acts as a safety-critical supervisor, guaranteeing team cohesion by enforcing a maximum human-robot separation distance.

This dual-layer protocol ensures that the human partner is implicitly informed of the robot team's capabilities, allowing for a fluid and safe partnership. The detailed mathematical formulation of this novel coordination protocol is presented in Section (5).

Problem Statement: The objective is to design a real-time motion planning framework that computes a sequence of optimal control inputs, $\mathbf{U}^*$, over a prediction horizon H . These inputs must steer the formation's Center of Mass, $\mathbf{p}_{\text{obj}}$, to follow a human partner. This is achieved by dynamically generating a reference trajectory, $\mathbf{p}_{\text{ref}}$, based on the human partner's perceived state. The solution must satisfy all robot kinematic constraints, maintain the rigid grasp formation, and ensure collision-free navigation. Critically, the framework must also simultaneously provide feedback to the human partner via an adaptive coordination protocol to guarantee robust and stable team cohesion in dynamic environments.

4. MPC-Based Motion Planning Framework

To address the human-guided cooperative transport problem, we propose a centralized, nonlinear Model Predictive Control (NMPC) framework, the architecture of which is illustrated in Fig. 2. The controller computes optimal velocity commands for the fleet in real-time by repeatedly solving a constrained finite-horizon optimal control problem at each time step k . This predictive approach allows the system to anticipate the motion of the human and obstacles, enabling proactive and smooth control actions.

The NMPC framework is structured as the following optimization problem. In this formulation, the collective system dynamics, denoted by F , are derived by applying the single-robot kinematic model f from Equation (3) to each of the N robots in the fleet. The vectors $\mathbf{Z}$ and $\mathbf{U}$ represent the concatenated state and control inputs for the entire formation, respectively.

$$\min_{\mathbf{U}_k} \sum_{j=0}^{H-1} \left(\|\mathbf{p}_{\text{obj}}[j] - \mathbf{p}_{\text{ref}}[j]\|_{\mathbf{W}_t}^2 + \sum_{i=1}^N J_{\text{grasp},i}[j] + \|\mathbf{U}[j]\|_{\mathbf{W}_u}^2 \right) + \|\mathbf{p}_{\text{obj}}[H] - \mathbf{p}_{\text{ref}}[H]\|_{\mathbf{W}_f}^2 \quad (17a)$$

$$\text{s.t. } \mathbf{Z}[j+1] = F(\mathbf{Z}[j], \mathbf{U}[j]), \quad \forall j \quad (17b)$$

$$\|\mathbf{p}_{\text{ee},i}[j] - \mathbf{p}_{\text{ee},l}[j]\|_2^2 - d_{il,\text{des}}^2 = 0, \quad \forall i, l, j \quad (17c)$$

$$\text{dist}(C_{\text{robot},i}[j], C_{\text{obs},m}[j]) \geq d_{\text{safe}}, \quad \forall i, m \in \mathcal{O}_s, j \quad (17d)$$

$$\mathbf{a}_{\text{vo},m}[j]^T (\mathbf{v}_{\text{obj}}[j] - \mathbf{v}_{\text{obs},m}[j]) \geq \epsilon, \quad \forall m \in \mathcal{O}_d, j \quad (17e)$$

$$\|\mathbf{u}_{b,i}[j]\|_2 \leq V_{\text{max}}, \quad \forall i, \forall j \quad (17f)$$

$$\mathbf{q}_{a,i,\text{min}} \leq \mathbf{q}_{a,i}[j] \leq \mathbf{q}_{a,i,\text{max}}, \quad \forall i, \forall j \quad (17g)$$

$$\mathbf{Z}[0] = \mathbf{Z}_{\text{current}} \quad (17h)$$

Here, the optimization is performed over a prediction horizon of H steps, indexed by $j \in \{0, \dots, H-1\}$. The indices $i, l \in \{1, \dots, N\}$ refer to individual robots, while m denotes an obstacle from the environment sets $\mathcal{O}_s$ and $\mathcal{O}_d$.

The components of this formulation are detailed below.

The Objective Function (17a) is a weighted sum of quadratic cost terms designed to achieve robust tracking and formation stability:

1. *Human Tracking Cost:* The term $\|\mathbf{p}_{\text{obj}}[j] - \mathbf{p}_{\text{ref}}[j]\|_{\mathbf{W}_t}^2$ penalizes the deviation between the predicted position of the formation's COM and the dynamically generated reference trajectory, $\mathbf{p}_{\text{ref}}$, which is derived from the human's perceived position. The positive semi-definite weight matrix $\mathbf{W}_t$ tunes the tracking aggressiveness. This is the primary driver for the team's overall motion.
2. *Grasp Maintenance Cost:* The term $\sum_{i=1}^N J_{\text{grasp},i}[j]$ is crucial for ensuring a firm grasp throughout the maneuver. For each robot i , this cost penalizes the deviation of its end-effector's predicted pose from the desired grasp pose on the object. The desired grasp position in the world frame, $\mathbf{p}_{\text{grasp},i}[j]$, is a function of the object's predicted state at that time step:

$$\mathbf{p}_{\text{grasp},i}[j] = \mathbf{p}_{\text{obj}}[j] + \mathbf{R}(\psi[j])\mathbf{g}_i \quad (18)$$

where $\mathbf{g}_i$ is the fixed grasp point in the object's frame. The cost is then formulated as:

$$J_{\text{grasp},i}[j] = \|\mathbf{p}_{\text{ee},i}[j] - \mathbf{p}_{\text{grasp},i}[j]\|_{\mathbf{Q}_p}^2 + \|\mathbf{R}_{\text{ee},i}[j] - \mathbf{R}(\psi[j])\|_{\mathbf{Q}_o}^2 \quad (19)$$

The predicted end-effector pose ($\mathbf{p}_{\text{ee},i}$, $\mathbf{R}_{\text{ee},i}$) is calculated within the NMPC using the Product of Exponentials (PoE) forward kinematics. This cost actively engages the manipulator arms to maintain the grasp.

3. *Control Regularization Cost:* The term $\|\mathbf{U}[j]\|_{\mathbf{W}_u}^2$ penalizes the magnitude of the control inputs for both the mobile bases and the arms. Its purpose is to ensure the generated velocity commands are smooth, continuous, and energy-efficient. $\mathbf{W}_u$ is a positive definite weight matrix.

4. *Terminal Cost*: The term $\|\mathbf{p}_{\text{obj}}[H] - \mathbf{p}_{\text{ref}}[H]\|_{\mathbf{W}_f}^2$ is a cost applied exclusively at the final prediction step, H . By heavily penalizing the terminal tracking error with a large weight matrix $\mathbf{W}_f$, it discourages myopic planning and steers the system towards a stable state.

System Dynamics (17b) This equality constraint ensures that the predicted trajectory adheres to the discrete-time kinematic model of the robot fleet, as defined in Section 3.1.

Formation Maintenance (17c) To maintain the rigid grasp, this hard equality constraint is imposed. It fixes the squared Euclidean distance between the end-effectors of every pair of robots (i, l) to its desired value $d_{il,\text{des}}^2$, ensuring the geometric integrity of the formation throughout the task.

Static Obstacle Avoidance (17d) This constraint prevents collisions with known static obstacles from the set $\mathcal{O}_s$. We model each robot i and each static obstacle m by their conservative circular bounding volumes, $C_{\text{robot},i}$ and $C_{\text{obs},m}$, respectively. The function $\text{dist}(\cdot, \cdot)$ computes the Euclidean distance between the centers of these volumes. This nonlinear inequality constraint ensures this distance is always greater than a minimum safety margin d_{safe} , creating a protective buffer around all entities. Constraint (17d) enforces the safety requirement defined in (12) over the prediction horizon, ensuring collision-free operation for each individual robot i .

Dynamic Obstacle Avoidance (17e) For moving obstacles from the set $\mathcal{O}_d$, a purely distance-based constraint is reactive and insufficient for smooth planning. We therefore employ the Velocity Obstacle (VO) method [29, 30]. The VO defines a cone in the relative velocity space containing all velocities that would lead to a future collision, assuming the obstacle maintains its current velocity. To guarantee safety, the planned relative velocity of the object must lie outside this collision cone. Constraint (17e) formulates this as a linear inequality, where $(\mathbf{v}_{\text{obj}}[j] - \mathbf{v}_{\text{obs},m}[j])$ is the predicted relative velocity, $\mathbf{a}_{\text{vo},m}[j]$ is the normal vector to a hyperplane that separates the safe velocities from the unsafe collision cone, and ϵ is a small positive scalar ensuring the velocity remains strictly on the safe side. This predictive constraint allows the NMPC to proactively compute smooth, collision-free escape maneuvers.

Physical Limitations (17f) - (17g) The hardware limitations are respected through state and control limits. These constraints bound the allowable base velocities and manipulator joint angles within their physical operational ranges.

Initial State (17h) The optimization is anchored to reality at each time step by this constraint, which requires the planned trajectory to start from the system's current measured state, $\mathbf{Z}_{\text{current}}$.

By solving this constrained optimization problem (17) at each time step, the framework generates optimal, safe, and coordinated control actions that allow the robotic team to robustly follow a human partner.

Hybrid Formation Control - For enhanced robustness against rare events such as large external disturbances or solver numerical issues, our implementation includes a simple, high-gain proportional recovery controller. If the formation error, measured at the end-effector level, exceeds a predefined tolerance, this controller momentarily overrides the NMPC to apply immediate corrective velocities to the bases, rapidly restoring the formation. Once the error is within tolerance, nominal NMPC operation seamlessly resumes. This practice ensures the integrity of the formation is maintained under all operating conditions.

5. Adaptive Human-Robot Coordination Protocol

While the NMPC framework in Section (4) computes optimal trajectories for the robot team, robust human-robot coupling requires a feedback mechanism to inform the human of the robot team's status. A naive implementation can lead to instability or unsafe separation if the human moves too quickly or into a complex area. To solve this, we introduce a novel, two-level adaptive coordination protocol that provides signals to the human based on the robot's real-time performance and is visualized in Fig. 4.

5.1. Proactive Feedback via Cost Monitoring

The first level acts as an anticipatory "early warning" system. It uses the NMPC's own internal cost, $J^*[k]$, as a powerful predictor of future task difficulty. If the system predicts a challenging maneuver (e.g., navigating a tight corner, avoiding multiple obstacles), the cost will rise. This triggers a signal for the human to slow down *before* significant tracking errors can accumulate. This is formalized as follows:

$$\text{Signal}_{\text{proactive}}[k] = \begin{cases} \text{'PROCEED'} & \text{if } J^*[k] < J_{\text{thresh}} \\ \text{'SLOW DOWN'} & \text{if } J^*[k] \geq J_{\text{thresh}} \end{cases} \quad (20)$$

where J_{thresh} is a tuned cost threshold. This allows a cooperative human to moderate their speed, reducing the control burden on the robot team.

5.2. Reactive Safety Supervision

The second level serves as a safety-critical supervisor that helps the team to remain spatially coherent, preventing any unsafe separation between human partner and the robot formation. It directly monitors the Euclidean distance between the human, $\mathbf{p}_h[k]$, and the formation's center, $\mathbf{p}_{\text{obj}}[k]$, denoted as $D_{hr}[k] = \|\mathbf{p}_h[k] - \mathbf{p}_{\text{obj}}[k]\|$.

To prevent oscillations, this level operates with hysteresis logic, using two distance thresholds: a maximum

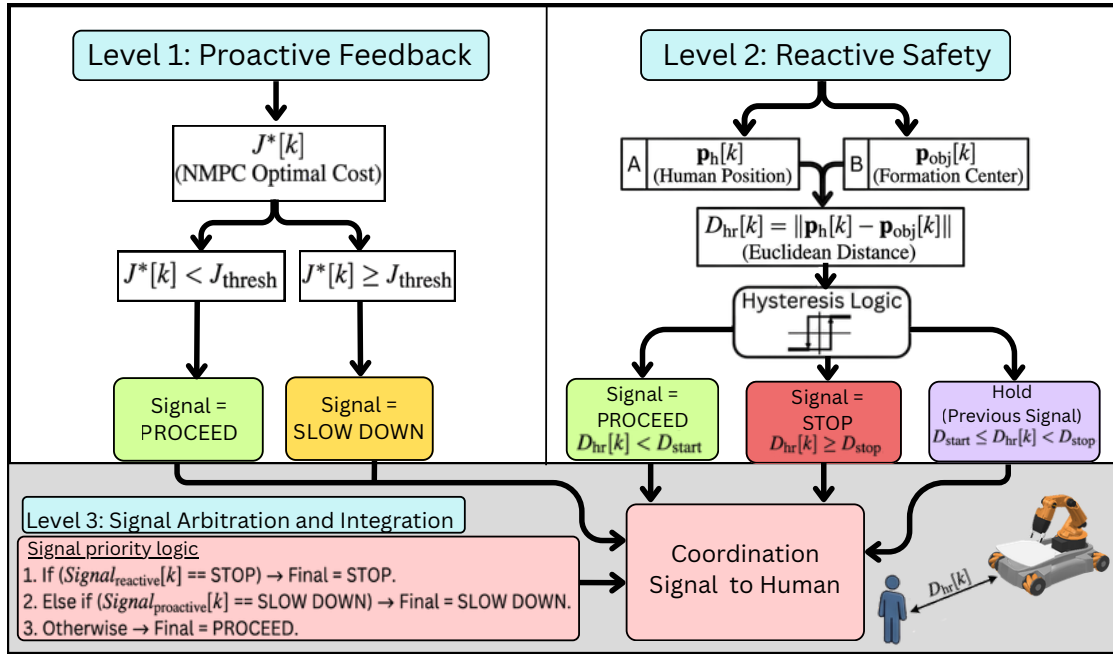


Figure 4: Adaptive Human-Robot Coordination Protocol. This diagram illustrates a two-level control strategy for human-robot interaction. Level 1 (Proactive Feedback) uses NMPC optimal cost $J^*[k]$ (Eq. 20) to generate 'PROCEED' or 'SLOW DOWN' signals, anticipating task difficulty. Level 2 (Reactive Safety) monitors the Euclidean distance $D_{hr}[k]$ between the human $\mathbf{p}_h[k]$ and formation center $\mathbf{p}_{\text{obj}}[k]$ (Eq. 21). It employs hysteresis logic to output 'PROCEED', 'STOP', or 'HOLD' signals, ensuring spatial coherence. These signals are then arbitrated based on a priority logic ('STOP' > 'SLOW DOWN' > 'PROCEED') to produce a unified Coordination Signal for the human, with a closed-loop feedback mechanism.

separation limit, D_{stop} , and a resumption threshold, D_{start} , where $D_{\text{start}} < D_{\text{stop}}$. This logic generates a critical signal as follows:

$$\text{Signal}_{\text{reactive}}[k] = \begin{cases} \text{'STOP'} & \text{if } D_{hr}[k] \geq D_{\text{stop}} \\ \text{previous signal} & \text{if } D_{\text{start}} \leq D_{hr}[k] < D_{\text{stop}} \\ \text{'PROCEED'} & \text{if } D_{hr}[k] < D_{\text{start}} \end{cases} \quad (21)$$

The 'STOP' signal is a critical request for the human to wait, ensuring the formation does not get left behind. Together, these two layers create a robust and intelligent coordination system where the human and robots can fluidly and safely collaborate on complex transport tasks.

5.3. Signal Arbitration and Integration

To provide the human partner with a single, unambiguous instruction, the outputs from the proactive and reactive coordination layers are integrated via a priority-based arbitration logic. As illustrated in the Level 3 architecture of Fig. 4, safety-critical requirements strictly override performance-related suggestions. The final coordination signal, $S_{\text{final}}[k]$, is determined by the following hierarchy:

$$S_{\text{final}}[k] = \begin{cases} \text{'STOP'} & \text{if } \text{Signal}_{\text{reac}} = \text{'STOP'} \\ \text{'SLOW DOWN'} & \text{else if } \text{Signal}_{\text{proac}} = \text{'SLOW DOWN'} \\ \text{'PROCEED'} & \text{otherwise} \end{cases}$$

This arbitration logic ensures that if the human-robot separation distance approaches the safety threshold D_{stop} , the system prioritizes a full halt, irrespective of the NMPC's current predictive cost. The 'PROCEED' signal is issued only when both the formation's spatial coherence is maintained and the predictive tracking performance remains within nominal bounds, thereby maintaining a fluid and safe collaborative pace.

Human-in-the-Loop Interaction: This framework provides feedback to the human partner by signaling the robot team's state. The proactive 'SLOW DOWN' signal allows a co-operative human to adapt their pace for smooth, continuous operation. The reactive 'STOP' signal provides a hard safety. If a human ignores the proactive signal and the team separates, the robots will physically slow down and eventually halt as their reference point stops updating relative to them, providing unmistakable physical feedback and ensuring team cohesion is never compromised. The system prioritizes safety by pausing the task rather than allowing formation degradation.

6. Experimental Validation

This section presents a rigorous experimental evaluation of the proposed NMPC framework. We performed simulation-based experiments aimed at assessing the performance of the system in producing dynamically feasible,

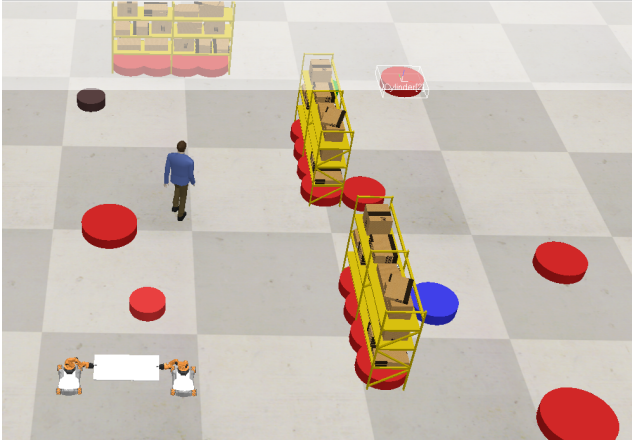


Figure 5: The experimental setup in the CoppeliaSim environment, showing the two-robot formation grasping a plate, the human partner, and a field of static and dynamic obstacles

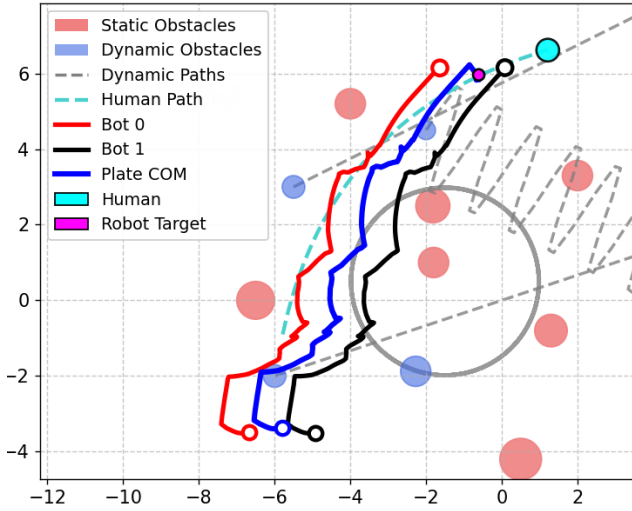


Figure 6: Executed trajectories from a representative simulation run, demonstrating the framework's ability to navigate complex and cluttered environment. The trajectory traced by the object's Center of Mass (blue) successfully tracks the dynamically generated reference path derived solely from the human's real-time position. The NMPC controller has no prior knowledge of the global trajectory and generates reference targets $\mathbf{p}_{\text{ref}}$ incrementally at each time step k . The cyan dashed line represents the actual trajectory taken by the human partner throughout the experiment, provided for visualization purposes only. The individual robot trajectories for Bot 0 (red) and Bot 1 (black) show the corrective maneuvers required to maintain the rigid formation while avoiding static and dynamic obstacles.

collision-free paths and, importantly, at testing the effectiveness of our coordination protocol in securing robust human-robot team cohesiveness.

6.1. Experimental Protocol

All experiments were conducted within a high-fidelity CoppeliaSim physics environment. The cooperative system,

shown in Fig. 5, comprises two holonomic KUKA youBot platforms transporting a rigid plate. A challenging testbed has been created by populating the workspace with both static and dynamic obstacles for evaluating the controller's predictive capabilities. The task required the robotic team to track a human partner agent along a complex, non-trivial reference path. The human's progression along this path was governed by the state-dependent coordination protocol detailed in Section 5. The NMPC framework was implemented in Python using CasADi with the IPOPT solver.

6.2. Performance Analysis and Results

The results validate the framework's capability to produce sophisticated, intelligent behavior while ensuring safety and stability. We analyze the performance of the system by examining the trajectory execution and the behavior of the multi-level coordination protocol. The quantitative performance metrics and task completion statistics across more than 50 simulation trials are summarized in Table 2.

6.2.1. Trajectory Tracking and Collision Avoidance

Fig. 6 provides a qualitative overview of a complete experimental run. The trajectory of the formation centroid (blue line) successfully follows the dynamically generated reference trajectory. The human's trajectory was not known a priori. The robot trajectories (red and black lines) illustrate the corrective maneuvers calculated by the NMPC to ensure the formation's integrity while executing complex turning maneuvers. The predictive nature of the controller is evident as the formation smoothly navigates around the static obstacles without any reactive, jerky motions.

6.2.2. Analysis of the Multi-Level Coordination Protocol

A critical contribution of this work is the multi-level protocol for ensuring robust human-robot coupling. We now analyze the performance of both its proactive and reactive layers. A *supplementary video demonstrating this behavior is available online¹, specifically highlighting the correlation (at video timestamp 1:04 - 1:24) where the robotic team struggles with obstacles and the human partner safely waits.*

Proactive Feedback via Cost Monitoring The first level of the protocol uses the NMPC's internal cost as a predictive metric for task difficulty. Fig. 7 analyzes this relationship. The plot shows the NMPC objective cost (red line) and the resulting human speed (blue line) over time. The human speed, $\|\mathbf{v}_h[k]\|$, is obtained by logging the agent's instantaneous velocity magnitude within the CoppeliaSim environment at each sampling step. Notably, the human speed dynamically adjusts in response to the proactive 'SLOW DOWN' and reactive 'STOP' signals, demonstrating a closed-loop human-in-the-loop interaction. During intervals where the controller anticipates high difficulty in tracking and obstacle avoidance (e.g., at $t \in [20, 120]\text{s}$), the cost value

¹Supplementary video and project abstract are available at: https://shruti0320.github.io/cooperative_object_transport/

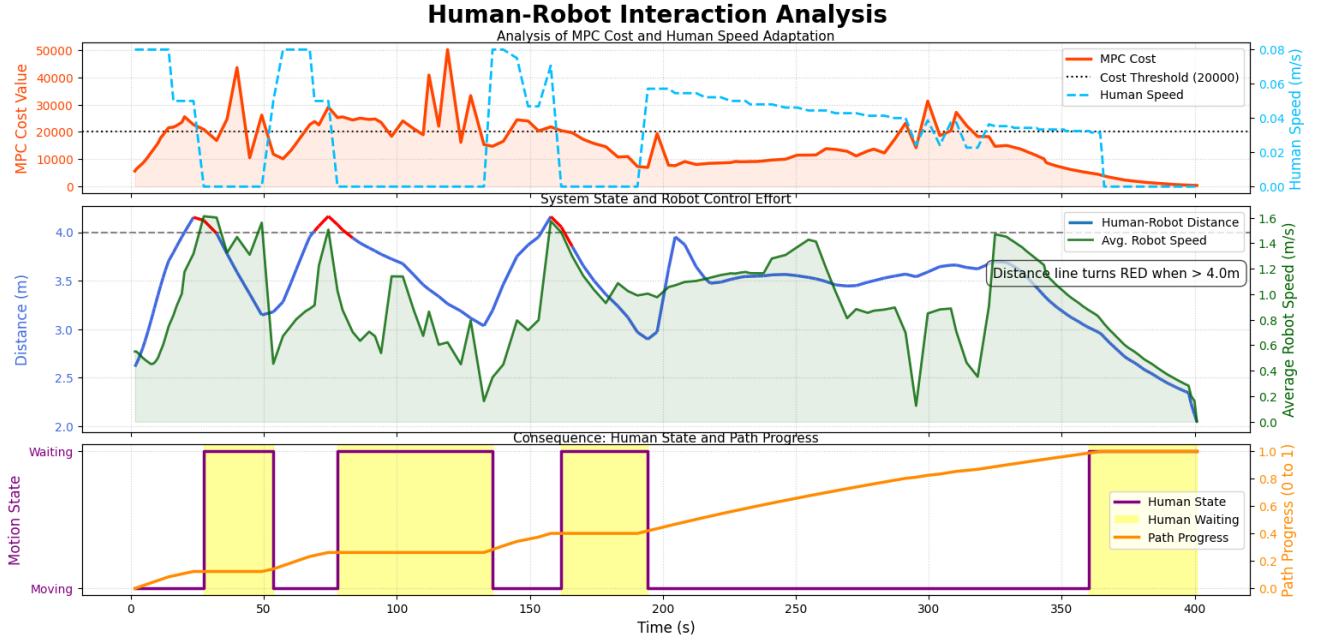


Figure 7: Comprehensive Analysis of Human-Robot Interaction Dynamics. This figure presents a three-tiered analysis of the interaction over time. (Top) The relationship between the robot's *MPC Cost* (red) and the human's adaptive *Human Speed* (blue), which modulates in response to the robot-generated coordination signals. The velocity data is logged directly from the simulation environment, representing the partner's cooperative response to proactive and reactive feedback. (Middle) The core feedback loop, illustrating the *Human-Robot Distance* (blue/red) and the responsive *Average Robot Speed* (green). The line turns red when the safety distance threshold is breached. (Bottom) The resulting consequences on human behavior, plotting the binary *Human State* (purple: 0 for Moving, 1 for Waiting) against the cumulative *Path Progress* (orange). The yellow shaded areas explicitly highlight periods where the human is waiting for the robots.

risers. When this cost surpasses the pre-defined proactive threshold (dashed line), the protocol intervenes, alerting the human to reduce its nominal speed. If the human does not reduce speed, the prolonged high-cost state triggers a more urgent alert. This anticipatory deceleration in human's speed lessens the control burden on the follower team before a significant tracking error can accumulate, validating the effectiveness of the proactive strategy.

Reactive Safety Supervision. The second level of the protocol acts as a safety-critical supervisor. Fig. 7 provides a quantitative analysis of this reactive mechanism. The middle part of the figure shows that when the separation distance between the human and the formation goes beyond the separation limit (D_{stop}), the protocol issues a 'STOP' signal. The bottom subplot reveals the consequence: the human's "Motion State" (purple line) transitions to a "Waiting" state and progression along the reference path is paused. This is visualized by the flattening of the 'Path Progress' metric (orange line). This mechanism ensures that the team remains spatially bounded and prevents human-robot decoupling under any condition, validating the protocol's effectiveness as a critical safety layer.

6.2.3. Ablation Study: Efficacy of the Coordination Protocol

To rigorously validate the necessity and effectiveness of our adaptive coordination protocol, we conducted a comparative ablation study. We compare the performance of our proposed method against a *naive follower* baseline under identical, challenging experimental conditions.

Baseline Definition- The naive follower utilizes the same core NMPC controller but operates in an open-loop, non-cooperative manner; the feedback loop to the human is severed. The human partner proceeds at their nominal pace, completely unaware of the robot team's status or the difficulties it may be facing. For this study, we have considered a realistic perception limit, emulating a sensor's maximum range. If the human-robot separation exceeds this "Visual Sensor Limit (5.5 m)", the robots lose their tracking reference, resulting in an irrecoverable task failure. The primary question this study answers is: "Is the core NMPC sufficient on its own, or is the explicit communication protocol essential for success?"

Performance Analysis- Figure 9 and Table 1 present the starkly different outcomes of the two approaches.

Human-led Cooperative Transport

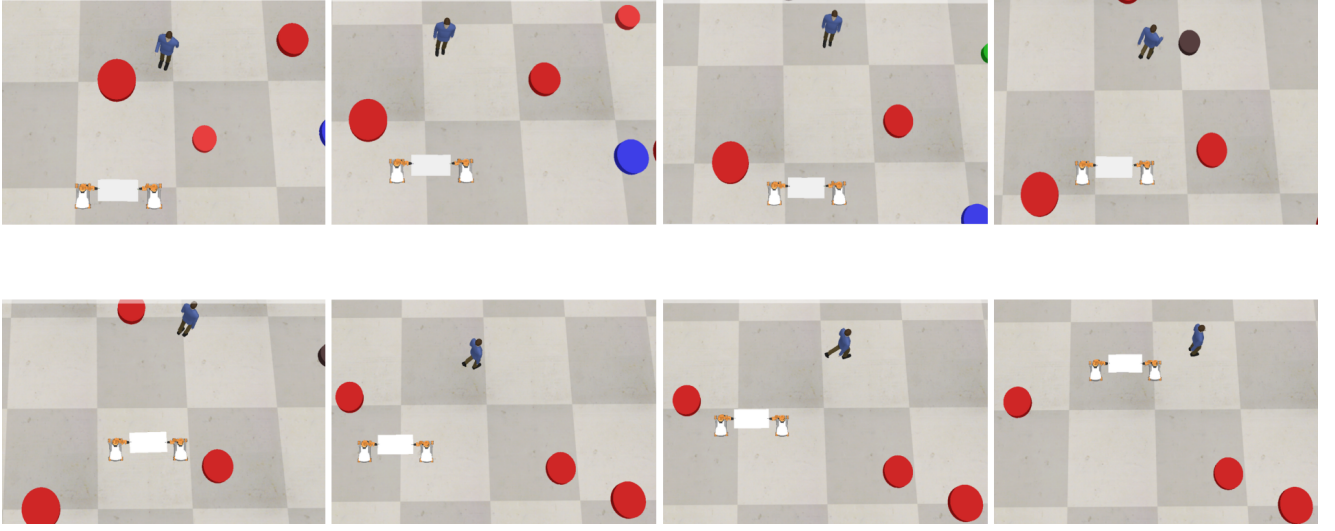


Figure 8: A time-series of snapshots from the simulation experiment. Two cooperative mobile robots transport an object while navigating a cluttered environment under the guidance of a human partner. The sequence demonstrates the system’s ability to maintain formation, follow the human partner and perform real-time collision avoidance.

Baseline Failure: Without the coordination protocol (red dashed line), the robot team is unable to keep pace with the human while navigating the environment’s challenges. The Human-Robot separation distance grows uncontrollably. It rapidly surpasses the reactive safety threshold (D_{stop}) and ultimately breaches the Visual Sensor Limit at approximately $t = 65$ s. At this point, the perception link is severed, the robots lose their tracking reference, and the mission results in a catastrophic failure with minimal path completion.

Proposed Protocol: In contrast, our proposed method (blue solid line) demonstrates robust, intelligent coordination. As the team encounters the same difficulties, the separation distance begins to increase. However, as it approaches the safety threshold (D_{stop}), the reactive ‘STOP’ signal is triggered. This is visually evidenced by the distance plot flattening out, remaining bounded well below the sensor limit. Guided by the protocol, the robotic team successfully signals the cooperative human to halt, allowing the formation to catch up and ensuring team cohesion is never compromised. As shown in Table 1, this strategy allows the system to maintain the perception link and achieve 100% task completion.

The results confirm that the feedback from the adaptive protocol is not merely beneficial, but is essential for successful task completion in scenarios where robot and human agility may differ.

7. Conclusion

This paper introduces a robust and smart framework that supports effortless human-guided cooperative transport by multiple mobile manipulators. Our system computes

Table 1

Quantitative Comparison: Baseline vs. Proposed Adaptive Protocol

Metric	No Coordination (Baseline)	Proposed Protocol (Ours)
Mission Outcome	FAILURE	SUCCESS
Visual Link Status	Broken ($t = 65.7$ s)	Maintained
Max Separation Distance	14.26 m	4.15 m
Path Completion	18.2% [†]	100.0%

[†] Path progress at the moment of failure (visual loss).

smooth, safe, and dynamically consistent trajectories in real time. This is done by defining the intricate interaction of system dynamics, task constraints, and environmental interactions as a single Nonlinear MPC.

Our core innovation is a hierarchical adaptive coordination protocol that allows a reliable negotiation between human intent and robotic capabilities. The forward-looking proactive layer acts as a ‘soft’ negotiation, intelligently slowing the task pace in anticipation of difficult maneuvers or high robotic effort. Concurrently, the reactive layer enforces a ‘hard’ safety boundary, guaranteeing team coherence by preventing the human-robot distance from exceeding a critical threshold.

Analysis of our experimental data which is visually summarized in Fig. 8, revealed a noteworthy emergent characteristic. The system implicitly negotiates between human objectives and robot-oriented constraints. The NMPC does not follow orders passively. Rather, it actively evaluates each command taking into account its safety and physical

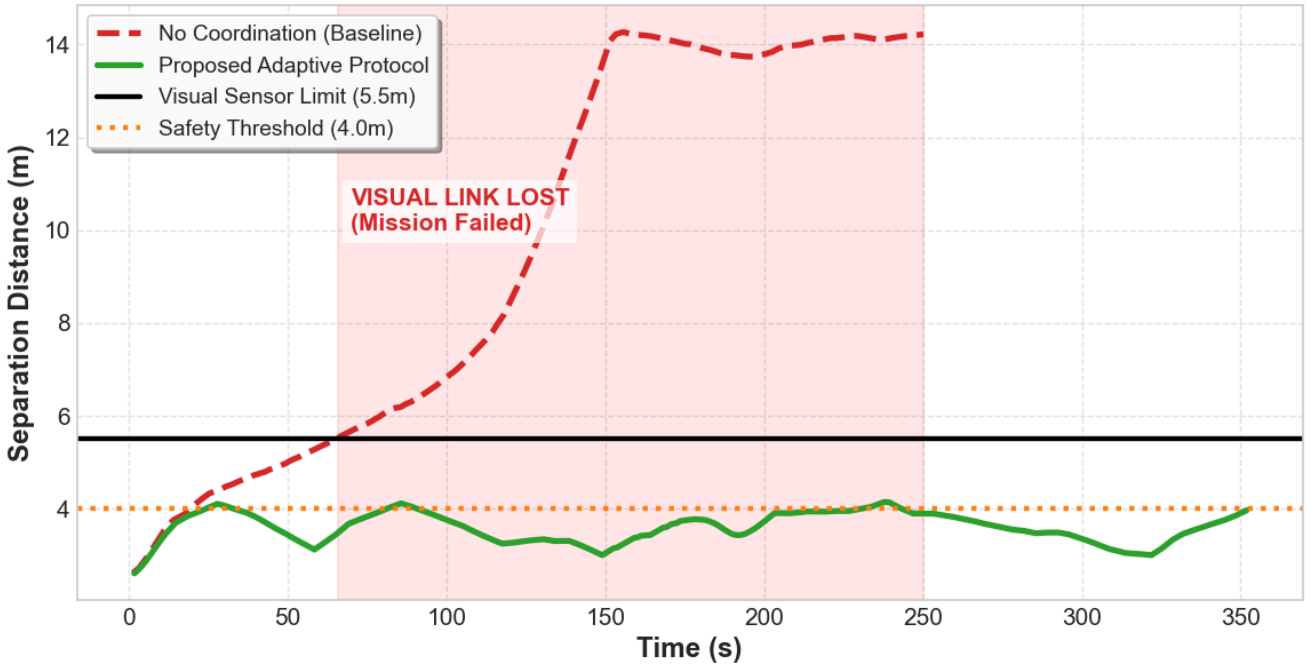


Figure 9: Comparative Ablation Study of Team Cohesion- This plot illustrates the Human-Robot separation distance over time. The *Naive Follower* (Red Dashed) demonstrates a catastrophic failure: without coordination, the robots fall behind, causing the separation to breach the Visual Sensor Limit (5.5m), resulting in mission failure. In contrast, the *Proposed Adaptive Protocol* (Green Solid) successfully regulates the separation distance, keeping it bounded by the reactive Safety Threshold (4.0m) and ensuring the mission is completed successfully.

constraints: Is it feasible?" Is it safe? When a conflict arises which is identified by the NMPC through high predictive costs or excessive separation distances, our multi-level coordination protocol is activated to resolve it by providing clear feedback to the human.

Our high-fidelity simulations validated this strategy across more than 50 trials, demonstrating emergent intelligent behaviors where the robotic team smoothly navigated challenging scenarios led by a human partner. This research represents a key step towards deploying human-centric robot teams in industrial and logistical settings. Future work will consist of decentralizing the control architecture for improved scalability and including improved human motion models to further close the gap between human intention and robotic execution.

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Table 2

Statistical Performance of the Proposed Protocol Across 50+ Trials

Performance Metric	Value
Task Completion Rate	96% [†]
<i>For Successful Runs:</i>	
Average Completion Time (s)	245.8 ± 15.2
Average Max. Separation (m)	4.05 ± 0.18
Average MPC Cost	19,500 ± 2,500

[†] The failures were attributed to IPOPT solver non-convergence under particularly dense obstacle configurations, resulting in no valid control output within the allotted computation time.

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